Differential Characteristics of the NIST Lightweight Cryptography Standard Ascon: 42 Active S-boxes over 4 Rounds and 68 over 5 Rounds

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Abstract

Ascon is the cryptographic algorithm that NIST standardised in 2025 to protect data on small devices such as IoT sensors. We report a 4-round differential characteristic of the Ascon permutation with 42 differentially active S-boxes and a 5-round characteristic with 68 active S-boxes. The best previously known characteristics had 43 and 72 active S-boxes, respectively, so the minimum number of active S-boxes over 4 rounds now lies between the proven lower bound of 36 and 42. Both characteristics were found with a bit-level SAT model in which the cardinality bound is tightened incrementally under assumptions. We verified every round of both characteristics against the designers' reference implementation and, in addition, with a machine-checked proof in Lean 4. The characteristics have probability 2^{-138} and 2^{-226} and do not threaten the security of Ascon.

Keywords: symmetric-key cryptography, cryptanalysis, differential attack, Ascon, NIST SP 800-232, SAT solver, Lean 4

1 Introduction

Ascon [1] is the family of authenticated encryption and hashing algorithms that NIST selected in 2023 for constrained devices and standardised in SP 800-232 [2]. Its 320-bit permutation acts on five 64-bit words $(x_0, \ldots, x_4)$. Each round adds a constant to x_2, applies a 5-bit S-box S to each of the 64 columns (with x_0 as the most significant bit), and then applies the linear layer

$$\Sigma_i(x_i) = x_i \oplus (x_i \ggg r_{i,1}) \oplus (x_i \ggg r_{i,2}), \tag{1}$$

with $(r_{i,1}, r_{i,2}) = (19, 28), (61, 39), (1, 6), (10, 17), (7, 41)$ for $i = 0, \ldots, 4$. An R-round differential characteristic is a sequence $(\Delta X_r, \Delta Y_r)_{r<R}$ of input and output differences of the S-box layer with $\Delta X_{r+1} = \Sigma(\Delta Y_r)$. Writing $a_{r,j}$ and $b_{r,j}$ for column j of ΔX_r and ΔY_r, and $\mathrm{DDT}[a][b] = \big|\{v \in \mathbb{F}_2^5 : S(v) \oplus S(v \oplus a) = b\}\big|$, the characteristic is valid if every $\mathrm{DDT}[a_{r,j}][b_{r,j}] > 0$, and its probability (under the usual independence assumption) and number of active S-boxes are

$$p = \prod_{r=0}^{R-1} \prod_{j=0}^{63} \frac{\mathrm{DDT}[a_{r,j}][b_{r,j}]}{32}, \qquad \# = \sum_{r=0}^{R-1} \big|\{\, j : a_{r,j} \neq 0 \,\}\big|. \tag{2}$$

We call $w = -\log_2 p$ the weight. A differential attack exploits an input difference that leads to a predictable output difference with high probability. Since every active S-box multiplies p by at most $8/32 = 2^{-2}$, the minimum of # over all non-trivial R-round characteristics is the standard yardstick for how many rounds resist such attacks.

For $R \leq 3$ the minimum is known: 1, 4 and 15 [1]. For $R = 4$ the designers reported a characteristic with 44 active S-boxes [1], Makarim and Rohit improved it to 43 [4], and Erlacher, Mendel and Eichlseder proved with SAT solvers that every 4-round characteristic has at least 36 [3]. For $R = 5$ the best known characteristic had 78 active S-boxes [1], improved to 72 [4]; the designers' heuristic search found none with fewer than 64. Makarim and Rohit note that a tight bound for 4 to 6 rounds is still not known [4]. El Hirch et al. [5] improved the related bounds on trail weight.

Contribution. We give a 4-round characteristic with 42 active S-boxes (pattern 15–11–9–7) and a 5-round characteristic with 68 active S-boxes (15–11–9–7–26). Table 1 summarises the state of knowledge.

Table 1: Minimum number of differentially active S-boxes of the Ascon permutation.

Rounds	proven lower bound	best known characteristic
3	15 [1]	15
4	36 [3]	44 [1] → 43 [4] → **42** (this note)
5	–	78 [1] → 72 [4] → **68** (this note)

2 Search method

Model. We use a bit-level SAT model of R rounds. Variables X_r and Y_r hold the input and output differences of the S-box layer in round r (5×64 bits each). For each column, the 10-bit relation "DDT$[a][b] > 0$" between input difference a and output difference b is encoded by blocking all impossible pairs; each blocking clause is shortened greedily while it still excludes no possible pair, which leaves 169 clauses per S-box (the DDT has 317 non-zero entries including $0 \to 0$). The linear layer (1) gives $X_{r+1} = \Sigma(Y_r)$ as 4-variable XOR constraints. An activity variable $A_{r,j}$ is forced to 1 whenever column j of X_r is non-zero. The round constants do not affect differences and are omitted. To remove the rotational symmetry of the state we require column 0 of X_0 to be non-zero.

Cardinality and Matsui-type constraints. The activity variables, in round order, feed one sequential counter [8] encoded in both directions, so that its register $c_{i,j}$ is true exactly when at least j of the first i variables are true. Following Sun, Wang and Wang [6], we add Matsui-type conditions [7]. With B_t a known lower bound for t rounds ($B_0 = 0$), the constraints are

$$\sum_{r=0}^{R-1}\sum_{j=0}^{63} A_{r,j} \leq k, \qquad \sum_{r=s}^{e}\sum_{j=0}^{63} A_{r,j} \leq k - B_s - B_{R-1-e} \quad (0 \leq s \leq e \leq R-1,\ (s,e) \neq (0,R-1)). \tag{3}$$

We used $B = (0, 1, 4, 15)$, and $B_4 = 36$ for five rounds. Each window constraint is expressed as implications $c_{\mathrm{end}(e),\,t+m+1} \Rightarrow c_{\mathrm{end}(s-1),\,t+1}$ between counter registers at round boundaries, where m is the right-hand side, $\mathrm{end}(r) = 64(r+1)$, $c_{0,t}$ is false for $t \geq 1$, and t ranges over $0 \leq t \leq k-m$.

Descending search. Asking directly for a 4-round characteristic with at most 44 active S-boxes did not return within one hour. Instead we build the model once for a loose bound k_0 (60 for four rounds) and, each time the solver returns a characteristic with a active S-boxes, solve again under the assumption $\neg c_{N,a}$, i.e. at most $a - 1$. The solver keeps its learned clauses between calls. With CaDiCaL 1.9.5 [9] through PySAT [10], on one core of an Apple M1 Max shared with other jobs, the sequence of characteristics found had 59, 58, 52, 49, 47, 46, 45 and

42 active S-boxes; the last one appeared after 2,821 s of wall-clock time. The solver jumped from 45 directly to 42. The model has 16,603 variables and 107,432 clauses. A second run with Glucose 4 reached 45 after 29 s and did not improve on it within the same time.

Five rounds. The 5-round characteristic extends the 4-round one forward by one round. Its last S-box layer has 7 active columns, each with 8 possible output differences; enumerating all 8^7 choices and applying Σ gives at least 26 active S-boxes in the fifth round, for a total of 68. Extending backward instead costs 57 S-boxes, because Σ^{-1} is dense.

3 The characteristics

Table 2 lists the 5-round characteristic. Its first four rounds form the 4-round characteristic with 42 active S-boxes. Words are written $x_0, \ldots, x_4$ in hexadecimal, bit 0 is the least significant bit, and S-box inputs are read column-wise with x_0 as the most significant bit, as in the specification.

Table 2: Differences at the input (ΔX_r) and output (ΔY_r) of the S-box layer. #: active S-boxes, w: $-\log_2$ of the probability of the round.

r	ΔX_r $(x_0, \ldots, x_4)$	ΔY_r $(x_0, \ldots, x_4)$	#	w
0	b9460e2000000000	0000000000000000	15	42
	0000004080000001	a000020080000000		
	39460c6080000000	39460c2000000001		
	0000000000000000	0000020000000000		
	0000000000000001	8000024000000000		
1	0000000000000000	0000000000000000	11	39
	a100120481400001	a100024480400001		
	a100120080000001	0000000000000000		
	0000020081000000	0000000000000000		
	a100024480400001	0000104001000000		
2	0000000000000000	0000000000000000	9	36
	2000906081020008	0000000000000000		
	0000000000000000	0000000000000000		
	0000000000000000	0000000000000000		
	2000906081020008	2000906081020008		
3	0000000000000000	0000004000000000	7	21
	0000000000000000	0000104044100440		
	0000000000000000	0000000000000000		
	0000000000000000	0000000000000000		
	0000104044100440	0000100044100440		
4	0000004000080400	0000000000080000	26	88
	8088b24ae4902660	0022082000000000		
	0000000000000000	0088a24880000220		
	0000000000000000	0088a20880000620		
	8022182264982440	8022182264982440		
		rounds 0–3 / rounds 0–4	42 / 68	138 / 226

4 Verification

Against the reference implementation. Independently of the SAT model, we checked each round with the designers' Python implementation `pyascon` [11]. For round r we build a state whose active columns take values v with $S(v) \oplus S(v \oplus a) = b$ for the column differences $a \to b$,

fill the other columns at random, apply one full round of the reference permutation (constant addition, S-box layer and linear layer) to the state and to the state XOR ΔX_r, and compare the output difference with ΔX_{r+1} (or with $\Sigma(\Delta Y_4)$ in the last round). All rounds matched in 20 random trials each. We also recovered the S-box table from the reference implementation and checked that it equals the table in the specification.

Machine-checked proof. We transcribed the permutation into Lean 4 (v4.33.1, core library only) from the specification: the S-box layer literally as its bitsliced description, the linear layer and the round constants. The column S-box and its DDT are derived from this definition rather than typed in. All statements below are proved by kernel evaluation (`decide +kernel`); no theorem uses `native_decide`, and each depends at most on the axiom `propext`. (i) The derived S-box equals the table of the specification. (ii) The 12-round permutation maps a fixed input to the output computed with `pyascon`, which ties the Lean definition to the reference implementation. (iii) Both characteristics are valid (every column transition has a non-zero DDT entry, consecutive rounds are linked by Σ, all words are 64-bit values), their first input difference is non-zero, and they have 42 and 68 active S-boxes with the per-round profiles given above. (iv) For every round r an explicit 320-bit state s_r satisfies

$$\mathrm{round}_r(s_r) \oplus \mathrm{round}_r(s_r \oplus \Delta X_r) = \begin{cases} \Delta X_{r+1} & r < R-1, \\ \Sigma(\Delta Y_{R-1}) & r = R-1, \end{cases} \tag{4}$$

where round_r is the full round function including its constant. These pairs realise each round separately; a single pair following all rounds would, as for any characteristic of this weight, require about 2^{138} trials. The whole proof is checked in about 12 s on an Apple M1 Max.

5 Discussion

The new characteristics narrow the 4-round interval for the minimum number of active S-boxes from $[36, 43]$ to $[36, 42]$ and lower the best known 5-round value from 72 to 68, which is still above the designers' heuristic threshold of 64. They say nothing about the security of Ascon: their probabilities, 2^{-138} and 2^{-226}, are far below 2^{-128}, the 4-round characteristic has a larger weight than the best known 4-round characteristic (weight 107 [5]), and the permutation has 12 rounds in its initialisation and finalisation. Fewer active S-boxes do not imply higher probability; the 43-S-box characteristic of [4] also had lower probability than the 44-S-box one.

We did not search for the minimum, and the SAT run for at most 41 active S-boxes over 4 rounds is still in progress.

The search code, the characteristics in machine-readable form, the verification script and the Lean proof are available at `https://github.com/otoinobuo/ascon-trails`.

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